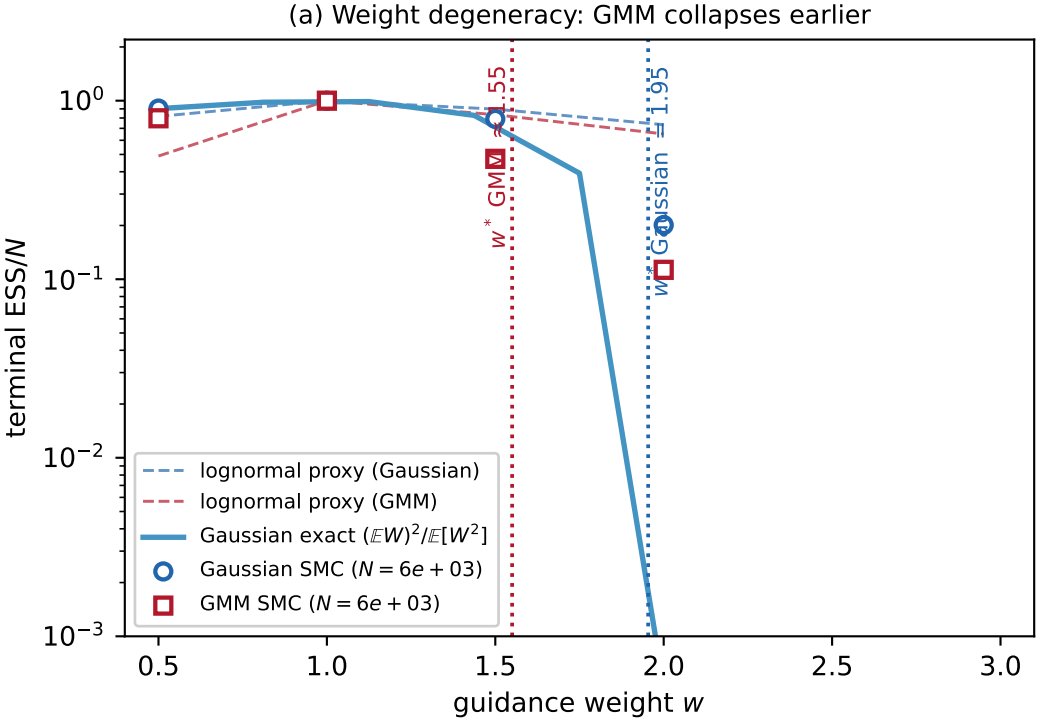
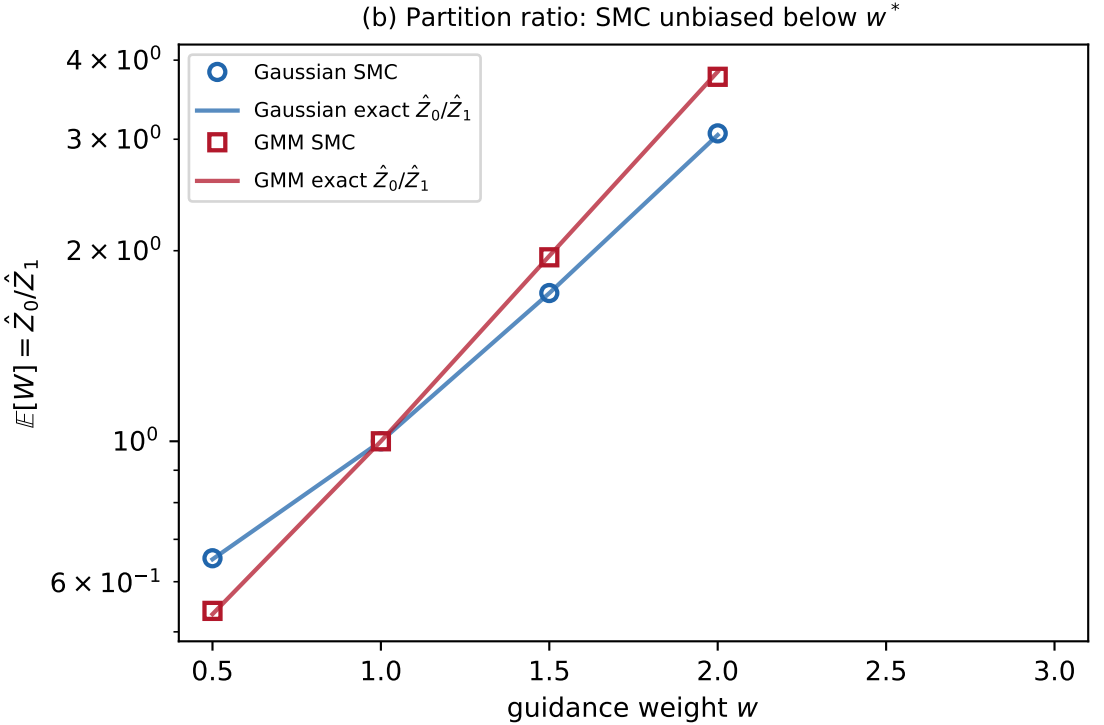

(a) Weight degeneracy: GMM collapses earlier
terminal ESS/N
guidance weight w
w^* GMM ≈ 1.55
w^* Gaussian ≈ 1.95
lognormal proxy (Gaussian)
lognormal proxy (GMM)
Gaussian exact $(\mathbb{E}W)^2/\mathbb{E}[W^2]$
Gaussian SMC ($N = 6e+03$)
GMM SMC ($N = 6e+03$)


(b) Partition ratio: SMC unbiased below w^*
$\mathbb{E}[W] = \hat{Z}_0/\hat{Z}_1$
guidance weight w
Gaussian SMC
Gaussian exact $\hat{Z}_0/\hat{Z}_1$
GMM SMC
GMM exact $\hat{Z}_0/\hat{Z}_1$